■ DAY 2 — NO MERCY CHALLENGE

Reframed: Lecture 2 Concepts ONLY

Available Concepts:
• String indexing & slicing
• String methods: len(), in, count(), index(), replace(), lower(), upper(), isupper(), islower(), isdigit(), isalpha()
• if, elif, else
• Comparison & Logical operators
• **NO LOOPS, NO LISTS, NO FUNCTIONS**

Q1 — Password Strength Check

A user enters a password. Check if it's **strong**:
• Must be at least 8 characters long
• First character must NOT be the last character
• Must contain at least one digit

Print "Strong" or "Weak".

Hint: Can you check if the string contains "0", "1", "2", etc. using the in operator?

Q2 — Palindrome (No Spaces)

A user enters a two-word phrase. Remove the space and check if it's a palindrome.

Example: User enters "race car" → becomes "racecar" → is it the same forwards and backwards?

Print "Yes" or "No".

Hint: Use .replace() to remove spaces, then use slicing to reverse.

Q3 — Email Address Check (Simple)

A user enters an email. Check if it has these properties:
• Contains exactly one @ symbol
• Has at least one character before the @
• Has at least one character after the @
• The character after @ is NOT a space

Print "Valid" or "Invalid".

Hint: Use .count() to count @ symbols and .index() to find its position.

Q4 — Check String Format

A user enters a string. Check if:
• The first character is an uppercase letter
• The last character is a digit
• The string is at least 6 characters long

Print "Matches" or "Does not match".

Q5 — Mirror Check (First Half vs Second Half)

A user enters a **4-character string**. Check if:
• Character at index 0 equals character at index 3
• Character at index 1 equals character at index 2

If the string is NOT 4 characters, print "Invalid length".
Otherwise print "Mirror" or "Not mirror".

Q6 — Text Analysis

A user enters a string. Print:
1. The first 3 characters
2. The last 2 characters
3. Characters at indices 1, 3, 5 (if they exist)
4. The string reversed

If the string is too short, print "N/A" for that part.

Q7 — Greeting Validator

A user enters text. Check if:
- It contains the word "hello" (case-insensitive)
- It contains at least one uppercase letter
- It has more than 5 characters

If ALL three are true, print "Valid greeting". Otherwise print "Invalid greeting".

Hint: Use .lower() with in for case-insensitive checking.

Q8 — Phone Number Format

A user enters a phone number as a string (e.g., "555-1234" or "555 1234"). Check if:
- It has exactly 8 characters (including the separator)
- Character at index 3 is either - or a space
- Characters at indices 0, 1, 2, 4, 5, 6, 7 are all digits

Print "Valid phone number" or "Invalid phone number".

Q9 — Username Checker

A user enters a username. Check if:
- It's between 5 and 12 characters (inclusive)
- It starts with a letter
- It ends with a digit
- It does NOT contain spaces

Print "Valid username" or "Invalid username".

Q10 — Secret Code Validator

A user enters a code (string). It's valid if:
- It has exactly 7 characters
- Characters at positions 0, 2, 4, 6 spell something meaningful (already exists)
- The character at index 3 is NOT the same as the character at index 5
- It contains at least one digit
- It does NOT contain the substring "BAD"

Print "Valid code" or "Invalid code".

Do NOT look at answers until you've attempted all questions! ■